Approach to the peripheral nervous system

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Objectives

Delineate	Delineate the clinically relevant peripheral nerve anatomy
Describe	Describe the common pathogenesis, clinical symptoms and signs of peripheral neuropathy.
Acquire	Acquire a clinical approach to assess the patient with symptoms and signs of peripheral neuropathy.
Understand	Understand the underlying taxonomy leading to the recognition of the most common peripheral neuropathies.

The peripheral nervous system (PNS)

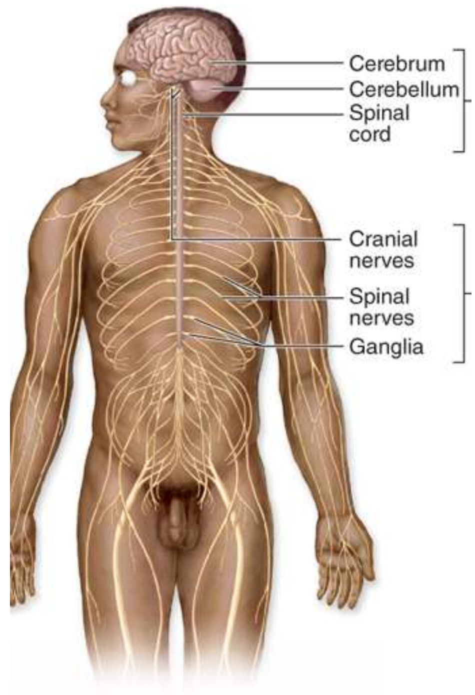
- Consists of motor, sensory, and autonomic neural elements with extensions outside of the brain and spinal cord
- **Motor neurons:** located in anterior gray matter of spinal cord, also called anterior horn cells.
  - Axons traverse nerve roots into peripheral nerves. Communicate with muscle at the neuromuscular junction
- **Sensory Neurons:** located in dorsal root ganglia outside the intervertebral foramina
- Disorders of the PNS may be **focal, multifocal, or generalized**
- Generalized disorders may be further divided into distal, proximal, symmetric, asymmetric, involving cranial or autonomic nerves

Anatomy

All the stuff outside the central nervous system!

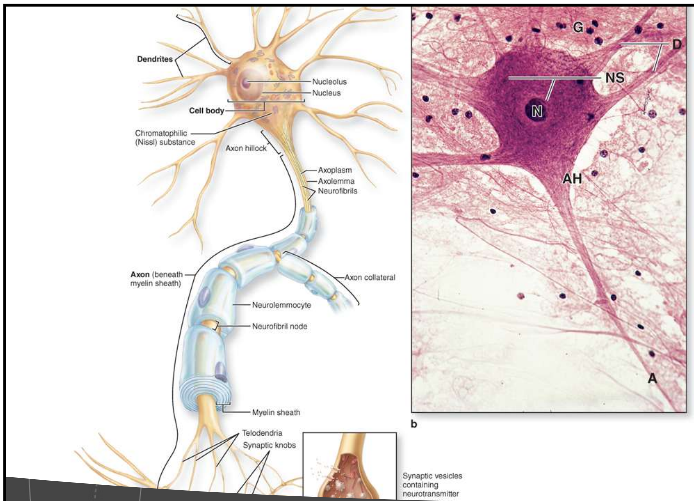
- Cranial Nerves
- Spinal Nerves
- Peripheral Ganglia
- Brachial Plexus
- Lumbosacral Plexus

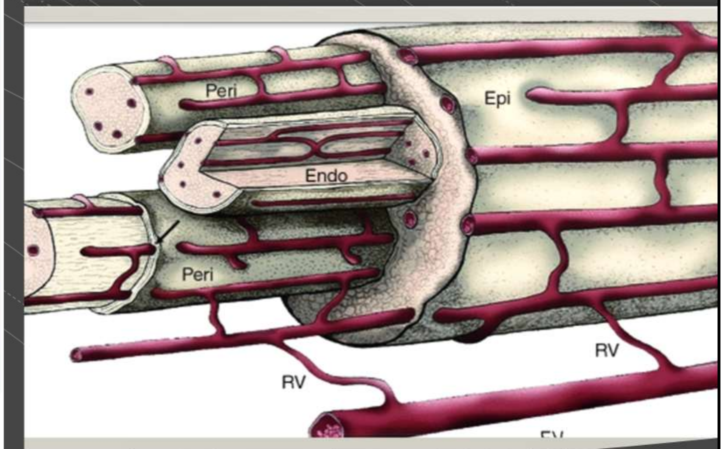
All are potential sites for pathology



Primary Sites of Pathology

- Axon (axonal neuropathy)
- Myelin (demyelinating neuropathy)
- Cell Body (Neuronopathy)
- Vasa nervorum
  - Vasculitic Neuropathies are often a manifestation of systemic autoimmune disease
  - Most often causes an axonal neuropathy





main sites of pathology

(a) A "typical" neuron has three major parts: (1) The **cell body** (also called the perikaryon or soma) is often large, with a large, euchromatic nucleus and well-developed nucleolus. The cytoplasmic contains basophilic **Nissl substance** or Nissl bodies, which are large masses of free polysomes and RER indicating the cell's high rate of protein synthesis. (2) Numerous short **dendrites** extend from the perikaryon, receiving input from other neurons. (3) A long **axon** carries impulses from the cell body and is covered by a myelin sheath composed of other cells. The ends of axons usually have many small branches (telodendria), each of which ends in a knob-like structure that forms part of a functional connection (synapse) with another neuron or other cell.

(b) Micrograph of a large motor neuron showing the large cell body and nucleus (**N**), a long axon (**A**) emerging from an axon hillock (**AH**), and several dendrites (**D**). Nissl substance (NS) can be seen throughout the cell body and cytoskeletal elements can be detected in the processes. Nuclei of scattered glial cells (**G**) are seen among the surrounding tissue. (X100; H&E)

Citation: Nerve Tissue & the Nervous System, Mescher AL. *Junqueira's Basic Histology: Text and Atlas, 15e;* 2018. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2430&sectionid=190279231 Accessed: September 28, 2019

Vasa nervorum are small arteries that provide blood supply to peripheral nerves. These vessels supply blood to interior parts of nerves and their coverings. Susceptible to external compression, target for vasculitic neuropathy, also implicated in diabetic neuropathy and others. From the surface, appears to lie parallel to the nerve, and then forms anastamoses throughout the nerve as seen in this slide. Form an extensive vascular network. Derived from the epineurium and reaches the endoneurium through the perineurium.

Epidemiology

- Neuropathy is very common – some estimates put prevalence of probable or definite polyneuropathy in patients over 80 years of age at >30%
- Evaluation of sensory disturbance, including peripheral neuropathy, is one of the 5 most common reasons for a neurologic consultation.
- Some very mild distal vibration loss is considered normal in the elderly, likely represents some mild degenerative neuropathic process
- **The lack of consensus about classification and variable terminology** contribute to the uncertainty about epidemiology

Approach to the patient

For a patient with sensory or motor symptoms, first determine:

- Does it localize to the CNS or Peripheral Nervous System?
- Can it localize to a single named Peripheral Nervous System structure?
- If it is Multifocal or generalized:
  - Does it involve sensory loss or weakness, or both?
  - Is it primarily proximal, distal, or both?
  - Is it symmetric or asymmetric?
- Is it acute or subacute?

Can't emphasize this enough, need to be able to confidently localize outside of the central nervous system. We can be doing a head-to-toe mri and emg/ncs on every patient with a neurologic complaint.

MORE THAN JUST REVLEXES!

Distinguishing central from peripheral nervous system Lesions

Distinguishing Central From Peripheral Nervous System Lesions

Features Consistent With Any Central Nervous System Localization

- Brisk reflexes in an affected limb
- Increased tone in an affected limb
- Sensory or motor symptoms limited to two limbs on the same side of the body
- Weakness of extensors in the upper extremity
- Weakness of flexors in the lower extremity

Features Consistent With a Spinal Cord Localization

- Sensory loss in both legs with a sensory level on the trunk
- Decreased pinprick sensation on one side, weakness and diminished proprioception on the other side

Features Consistent With a Peripheral Localization

- Axial or limb pain
- Diminished reflexes in an affected limb
- Weakness and diminished pain sensation in the same limb are not due to a spinal cord lesion and are due to either a peripheral or brain/brainstem lesion
- Sensory loss that involves the hands and feet but not the trunk

- CNS lesions don’t cause pain (with a few exceptions).
- Pain with neuropathic features (burning, tingling, shock like sensation) usually due to PNS lesions.
- Hyporeflexia in a symptomatic limb suggests PNS lesion

Common Presenting signs and symptoms

- Sensory Changes
- Sensory Ataxia
- Muscle Atrophy
- Autonomic Dysfunction
- Trophic Changes

These are the primary manifestations of Peripheral Nerve Disease. Some also list fasciculations, spasms and cramps, which can occur but are much less common and therefore not covered in this section.

Positive vs Negative Symptoms

- Important for further differentiation of type of pathology and management
- Positive symptoms include tingling, pain, buzzing, burning
- Negative symptoms include numbness, weakness, reflex loss

Most polyneuropathies cause impairment of motor and sensory function, but one often more affected than the other. Toxic and metabolic neuropathies, sensory loss exceeds weakness.

Axonal : sensation affected symmetrically in distal segments of the limbs and more legs than arms, owing to length-dependent nature of most disease that affect peripheral nerves.

Paresthesia and Dysesthesia

- Paresthesia and Dysesthesia are umbrella terms for abnormal sensations. Conventionally, paresthesia refers more to tingling type sensations and dysesthesia to painful sensation.
- Allodynia is pain caused by a stimulus that does not normally elicit pain.
- Hyperalgesia is an exaggerated response to a normally painful stimulus

Painful paresthesias and dysesthesias particularly common in diabetic, alcoholic-nutritional and amyloid neuropathies. They mainly affect feet ("burhing) and less often the hands.

Herpez zoster, shingles, dermatomal distribution, reactivation of virus in the dorsal root ganglion, inflammation can retrograde spread to spinal cord and involve motor system at or opposite to initial area of involvement. An unusual cause of "amyotrophy" or motor neuronopathy.

Sensory ataxia

- Proprioception is joint position sense
- Remember that it is proprioception that is feeding into the cerebellum for motor control, error correction, motor planning.
- Loss of proprioception in the lower extremities can cause ataxic gait – wide based, unsteady, unable to tandem walk
- Patients will have a positive Romberg (increased sway with eyes closed)

Reflex Changes

- One of the main indicators of lower motor neuron involvement
- Muscle stretch reflexes are frequently affected by disease of the peripheral nervous system
- The afferent limb of the muscle stretch reflex is carried by large myelinated axons to the CNS
- When strength is relatively preserved, depressed reflexes typically indicate involvement of the afferent limb
- With profound weakness, or neuromuscular junction localization, reflexes may be reduced as well.

Muscle Atrophy

- Shrinkage, wasting of muscle tissue
- A chronic finding, indicating either motor fiber component of peripheral nerve or lower motor neuron disease.
- Maximal degree of denervation atrophy after acute injury to axons occurs in 90-120 days (bulk ↓ 75-80%)
- Demyelinating neuropathies relatively spare muscle bulk due to lack of denervation.
- Look for patterns of atrophy to determine localization of peripheral nerve injuries
  - E.g.: thenar muscle wasting in severe carpal tunnel syndrome

Depending on the site of injury in a nerve, reinnervation by terminal sprouting of intact nerve terminals at the neuromuscular junction to denervated motor units can occur faster than healing of the nerve at the site of injury.

Autonomic Dysfunction

- Anhidrosis and orthostatic hypotension most common manifestations of Autonomic Dysfunction
- Less common:
  - small or medium size, unreactive pupils
  - lack tears and saliva
  - Erectile dysfunction
  - weak bowel/bladder sphincter
  - dilation of colon and esophagus

Most common in certain polyneuropathies, including amyloidosis or other small-fiber polyneuropathies especially DM.

In any neuropathy involving sensory nerves, there is loss of autonomic function in the distribution of the sensory loss. This is not true for radicular diseases because the autonomic fibers join the spinal nerves from the sympathetic chain and parasympathetic ganglia more distally.

Trophic Changes

- Denervation results in abnormal growth and/or atrophy of the innervated structures
- This can take the form of chronic sensory loss causing joint deformities (such as Charcot Neuropathic Osteoarthropathy)
- Autonomic denervation can lead to loss of hair growth, tight, and erythematous skin.
- More commonly seen in long term neuropathies, especially hereditary neuropathies (such as CMT, causing pes cavus, discussed later)

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Examination Techniques

- Motor Examination: Inspection of bulk, tone, strength
- Sensory Examination: Pin Prick, Light touch (cotton swab; Semmes-Weinstein monofilament), Tuning fork (128 Hz), Temperature
- Deep Tendon (myotatic) reflexes: Biceps, triceps, brachioradialis, patella, Achilles
- Trophic Examination: Skin color, texture, hair growth, sweating, temperature
- Gait and coordination testing
- Autonomic Testing (i.e.: Orthostatic blood pressure)

Monofilament and vibration most sensitive and specific than superficial pain (pinprick) or ankle reflex testing.

Single modality testing may miss 25-50% of those with DM PN.

Vibration and monofilament sensitive (90%) and specific (85-90%) screen for DMPN and correlates with foot ulcers.

Age related decline in vibration- 125% >65 y/o and 1/3rd of those older than 75 y/o have absent vibration on clinical examination.

Reduced vibration in isolation should not be over interpreted in the elderly or should prompt further history and consideration of another disorder.

Electrophysiologic Investigations

- Electromyography (EMG) and Nerve Conduction Study (NCS):
  - Commonly used in evaluation of peripheral neurologic complaints
  - Able to localize precise distribution of structures involved in certain circumstances (small fiber neuropathies may not show any abnormalities)
  - Able to assess the **post ganglionic fibers:**
    - sensory fibers: from the DRG and distally
    - motor fibers: from the anterior horn cells and distally

Nerve Conduction Study

- NCS utilizes electrical stimulus to measure speed and quality of action potentials down a nerve
- **Sensory Nerve Action Potential** – signal recorded over a distal or proximal site over a nerve, represents collective electrical activity of the largest myelinated sensory nerves.
- **Compound muscle action potential** – signal recorded over a distal muscle which represents the combined electrical activity produced in the muscle from a nerve
- **H reflex** – electrical equivalent of muscle stretch reflex (first medical experiement performed in the ISS!)
- **F waves** – backwards conducted (antidromic) signals up the nerves cause stimulation of anterior horn cells and result in f-waves

Electromyography

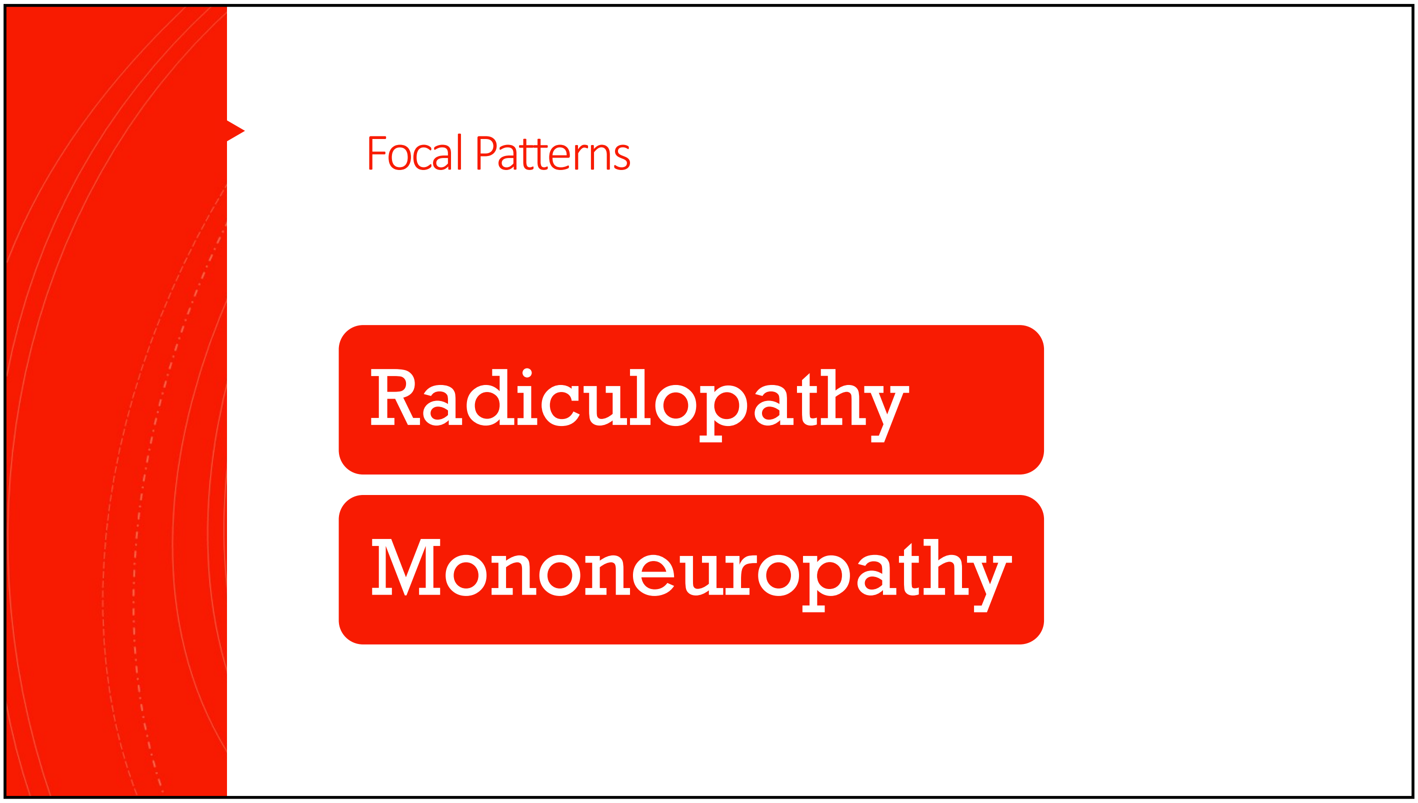
- EMG utilizes a needle electrode within a muscle for evaluation of muscle unit action potentials
- Needle electrode is placed in a muscle and electrical activity measured
- This is viewed on something like an oscilloscope
- **Motor Unit Action Potentials** are the signal of interest. Important aspects are recruitment pattern and insertional activity
- Able to provide better localization (which myotome/peripheral nerve distribution is involved)
- Can show evidence of *old* injury as well (reinnervation potentials)

Nerve Biopsy

- Not frequently utilized in the workup of common neuropathies
- May be necessary in vasculitic neuropathies (mononeuritis multiplex)
- Usually, the sural nerve is utilized as this is a pure cutaneous nerve and its area of innervation has minimal impact on function
- However, the diagnostic yield is low and there is a high risk of non-specific findings

Patterns of Involvement

Focal patterns vs Multifocal patterns



Polyneuropathy: multiple nerves, term often refers to bilateral, symmetric distribution

Radiculopathy: spinal nerve root versus polyradiculopathy, multiple spinal nerve roots. Generally a combination of motor and sensory impairment since ventral and dorsal roots are involved.

Neuronopathy or disease of the soma or neuron. Can be located in the ventral horn (anterior horn cells), the autonomic ganglia, the dorsal root ganglia or any combination depending on the pathologic process.

Mononeuropathy: single nerve

Muntiple mononeuropathies also called mononeuropathy multiplex. Would appear as though there are multiple nerves in disparate distributions, that are involved.

Plexopathy: diseases affecting the proximal plexi, and clinically we most often see brachial or lumbosacral plexus involvement.

Radiculopathy

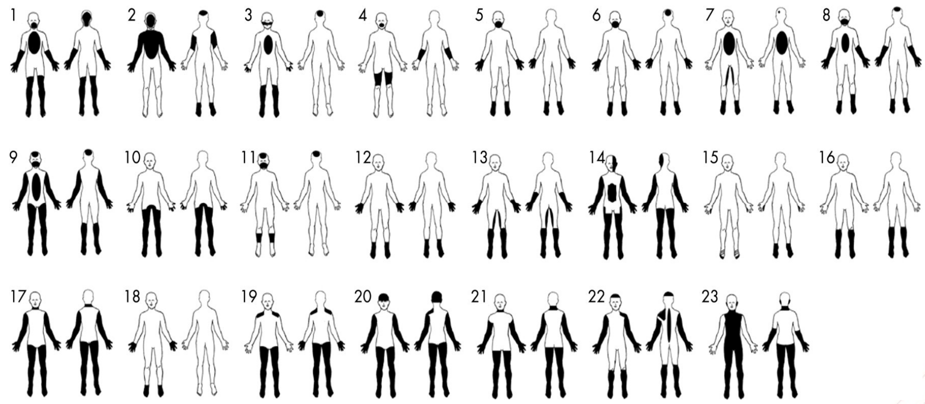
- **Site of pathology**: **nerve root**
- Usually due to compression of the root at the neuroforamen by a herniated intervertebral disc or by degenerative arthritic changes within the spine.
- Often presents as an acute or subacute onset of radiating pain **in the dermatome supplied** by the compressed nerve
- If compression severe enough, there may be loss of reflexes, loss of sensation, and weakness
- If chronic and untreated, may lead to significant muscle atrophy and disability
- Often self limited but if there is motor involvement, surgical evaluation is indicated

Mononeuropathy

- **Site of pathology: anywhere along a single peripheral nerve**
- Most circumscribed form of peripheral neuropathy
- Weakness and sensory loss within distribution of single nerve
- Difficult to differentiate from radiculopathy in some cases (L5 vs peroneal neuropathy)
- Entrapment Neuropathies:
  - Carpal tunnel syndrome, cubital tunnel syndrome, peroneal neuropathy at fibular head
- Cranial mononeuropathies also occur such as Bell's Palsy, a mononeuropathy involving the seventh cranial nerve

Sometimes can be difficult to differentiate a mononeuropathy from a single polyradiculopathy. The most common example is weakness in the muscles innervated by the L5 nerve root versus a mononeuropathy of the peroneal nerve. The latter usually associated with pain, weakness of tibialis anterior, tibialis posterior and peroneal muscles (so foot drop, weakness of ankle inversion and eversion) and sensory loss up to the knee. The former, mononeuropathy, would have more circumscribed sensory loss over the dorsum of the foot, and spares the tibialis posterior (preserves function of ankle eversion).

Multifocal patterns



Multifocal Patterns – a bit more complicated

- Many frameworks available. We will stick with this one:
- Three Axes:
  - Sensory, Motor, or Sensory-motor
  - Proximal, Distal, or Proximal + Distal
  - Symmetric and Asymmetric
- Most types of important multifocal neuropathic processes will fall into these categorizations
- Despite this good framework, you will hear terms on your clinical rotations that don't fit. This is a good starting place.
- You do not need to memorize the conditions on the following list, this is only for a 'big picture view.'

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Multifocal Patterns

Three axis framework for multifocal peripheral nervous disease

- We will cover some of these
- It is not an exhaustive list
- Do not memorize this list

	Sensory and Motor	Motor	Sensory
Proximal	Symmetric Diabetic lumbosacral radiculoplexus neuropathy Asymmetric	Symmetric Spinal bulbar muscular atrophy, brachial amyotrophic diplegia, most myopathies Asymmetric	Symmetric Asymmetric
Proximal and Distal	Symmetric AIDP, CIDP Mononeuritis multiplex, MADSAM, HNPP Asymmetric	Symmetric SMA, PMA, infectious/ paraneoplastic MND, some myopathies (ie, FSHD), NMJ disease, pure motor CIDP ALS, leg amyotrophic diplegia, MMN, some myopathies (ie, IBM) Asymmetric	Symmetric Dorsal column dysfunction, CISP Sensory neuronopathy Asymmetric
Distal	Symmetric Distal symmetric polyneuropathy, CMT, DADS Asymmetric	Symmetric Distal hereditary motor neuropathies, distal myopathies (ie, DM1), MMN Hirayama disease Asymmetric	Symmetric Distal symmetric polyneuropathy Asymmetric

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Proximal Asymmetric, sensory-motor

Diabetic Lumbosacral Radiculoplexus Neuropathy

- AKA Diabetic Amyotrophy
- One limb involved
- Usually painful combination of sensory impairment and weakness.
- Impairments do not correspond to distribution of adjacent nerve roots or well defined multiple single nerve distributions
- Reliance on imaging and EMG/NCV for diagnosis.
- Painful subacute onset of weakness and atrophy, usually involving the proximal lower extremity
- Risk does not correlate with severity of diabetes

Proximal Asymmetric, sensory-motor

Plexopathies

- Other considerations with similar presentation to diabetic amyotrophy:
  - Radiation plexopathy
  - Metastatic disease
  - Idiopathic brachial plexopathy

Proximal and Distal symmetric, sensory-motor

Acute Inflammatory Demyelinating Polyradiculoneuropathy

- AIDP is also called Guillain-Barre syndrome
- Starts with acroparasthesia, then ascending weakness and areflexia
- Develops over 2-3 weeks, hits peak by 4 weeks
- Can impair respiratory muscles – life threatening
- Often preceded by a self-limited infection such as Campylobacter Jejuni (know this for boards)
- We will come back to this, it is very important condition

Proximal and Distal symmetric, sensory-motor

Chronic Inflammatory Demyelinating Polyradiculoneuropathy

- CIDP shares some commonalities with AIDP
- Progression is slower, usually over about 8 weeks
- Does not typically cause respiratory or cranial neuropathies
- Dx with emg/ncs, lumbar puncture with albumino-cytologic dissociation (covered later in AIDP)
- Tx with IVIg, plasma exchange, or corticosteroids

Proximal and Distal Asymmetric, sensory-motor

Mononeuritis Multiplex

- Damage to at least two named peripheral nerves, most often not at entrapment sites
- Subacute (days– weeks) in onset
- Vasculitis is most common cause, destruction of the vasa nervorum -> ischemic nerve injury
- Usually associated with a systemic vasculitis:
  - Polyangiitis, granulomatosis with polyangiitis, microscopic polyangiitis, or polyarteritis nodosa.
  - Rhematoid arthritis, Sjögren syndrome, SLE
  - Lymphoma, HIV
  - Rarely caused by an isolated vasculitis of the PNS

Consider Hereditary Liability to Pressure Palsies if primarily at entrapment sites

Distal Symmetric, Sensory > Motor

Distal Symmetric Polyneuropathy

- Most common form of neuropathy
- This is what most people are referring to when they say neuropathy, polyneuropathy, etc.
- Most often sensory predominant, with motor manifestation late in course and variable with underlying cause
- Pain, paresthesia, and sensory loss in a length-dependent distribution
- Most common underlying etiology is diabetes, prediabetes, and unfortunately idiopathic disease
- Workup includes testing for DM, B12 deficiency, paraproteinemia (monoclonal gammopathy)
- Toxic etiologies should be considered in patients with history of chemotherapy or alcoholism

Distal Symmetric, Motor > Sensory

Distal Symmetric Polyneuropathy

- When motor symptoms are prominent, consider heritable forms of neuropathy
- Charcot-Marie-Tooth disease (CMT) is a family of hereditary motor and sensory neuropathies with variable age of onset, severity, and inheritance patterns
- Early neuropathic dysfunction leads to characteristic findings such as high arches and hammer toes
- Most common cause is duplication of PMP22 on chromosome 17 with autosomal dominant inheritance (CMT1) – demyelinating rather than axonal

Proximal and Distal, Symmetric, Pure Motor

Rarely due to a neuropathy

- Spinal muscular atrophy (discussed in the MND lecture)
- Consider non-neuropathic causes first such as neuromuscular junction or myopathic etiologies:
  - Myasthenia Gravis
  - Lambert Eaton Myasthenic Syndrome
  - Myopathies
  - Botulism – classic *descending* paralysis
- Rarely CIDP

Proximal and Distal, Asymmetric, Pure Motor

Motor Neuron Disease (motor *neuron*opathy)

- ALS discussed in detail in the motor neuron disease lecture.
- Infectious etiologies should be considered in acute-subacute settings
  - West Nile Virus
  - Enterovirus D68
  - Poliomyelitis is the prototypical infectious motor neuronopathy, though it is nearly eradicated (176 cases worldwide in 2019)

Distal, Symmetric, Pure Motor

Hereditary Motor Neuropathy

- Again, think CMT in this scenario
- Myotonic dystrophy type 1 (**not a neuropathy**)
  - Myotonia is delayed relaxation of muscles
  - Extensor digitorum brevis usually spared in distal myopathies but affected in distal symmetric sensorimotor neuropathies.
  - Due to *CTG* trinucleotide repeat expansion in DMPK gene

Distal, Symmetric, Pure Sensory

Small fiber neuropathies

- Characterized by painful paresthesia, dysesthesia, and atrophic changes of the skin
- Called ‘small fiber’ as it is the small, unmyelinated fibers which are dysfunctional in this condition
- Most common etiologies are diabetes, prediabetes, alcoholism, and paraproteinemia
- Frequently idiopathic

Specific Etiologies

Selected focal and multifocal neuropathies

Carpal Tunnel Syndrome

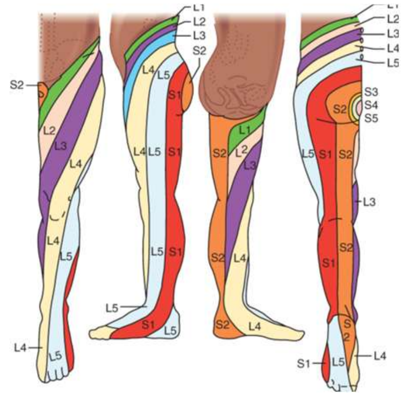
- Compression of the median nerve at the carpal tunnel (within the wrist.)
  - median nerve provides cutaneous sensation on palmar surface of thumb and index finger, as well as thumb abduction muscles
- Symptoms: painful tingling sensations in thumb and first few fingers - can have symptoms throughout hand and even up forearm.
  - "hand falls asleep," or "I wake up with pain and numbness in my hand."
  - "shaking" out the hand provides some relief

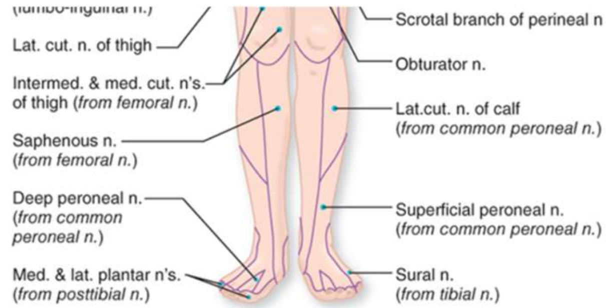
Carpal Tunnel Syndrome

- Evaluation: sensory exam, assessment for weakness in the thenar muscles
  - Tinel's sign over the wrist, phalen sign with flexion of the wrist
- Conservative treatment: soft wrist braces worn at night or during activities that provoke symptoms
  - More invasive procedures include corticosteroid injection and carpal tunnel release surgery
- If untreated, can lead to permanent thumb opposition weakness

Peroneal Neuropathy

- Common cause of a **'foot drop,'** or weakness of ankle flexion
  - The common peroneal nerve *or* deep peroneal nerve (a branch,) can cause foot drop, but **not the superficial peroneal nerve.**
- Frequently compressed at the fibular head
- Responsible for innervation of the **tibialis anterior**
- Tibialis anterior is supplied primarily by the L5 nerve root, making it difficult to distinguish between a radiculopathy at L5 and a mononeuropathy of the peroneal nerve.





Peroneal Neuropathy

- Notice the overlap of the sensory distributions between the L5 dermatome and the peroneal branches
- Compression at the common peroneal nerve or L5 nerve root can both cause foot drop, so how do we tell the difference?

Peroneal Neuropathy

- **Foot inversion** should be explicitly tested when a foot drop is evaluated, as this movement is accomplished by contraction of **tibialis posterior**, a muscle innervated by the tibial nerve.
- A common peroneal or deep peroneal nerve lesion should not cause much ankle inversion weakness, whereas an L5 radiculopathy would affect both muscles

Bell's Palsy

- Idiopathic condition resulting in paralysis of the facial nerve
  - Thought to occur due to compression of the nerve as it swells within the genu of the pathway through the temporal bone
- Unclear inciting cause of inflammation within nerve, several theories include autoimmune and viral causes
- Important to distinguish from stroke!
  - A true facial nerve palsy will **not spare the forehead or eye closure muscles** like an upper motor neuron injury

Bell's Palsy

- Depending on where along the facial nerve the injury is occurring, can also affect taste sensation in the ipsilateral anterior 2/3 of tongue and hyperacusis.
- Bell's phenomenon – same guy, different thing. When the eyes close, globes deviate upward. This is responsible for eye blink signals on EEG (will learn about in later lecture.)
  - In Bell's palsy, Bell's phenomenon can sometimes be seen as affected side has weakness of eye closure and globe still deviates up.
- Treatment: short course of corticosteroids. Several trials have assessed role for antivirals but have not shown benefit.
  - Antivirals should be prescribed if there is any concern about a viral etiology (fevers, rash, etc.)

AIDP or Guillain-Barre Syndrome

- Most common acute to subacute cause of generalized paralysis
- Effects all ages, genders, global, nonseasonal and nonepidemic (in general)
- 60% preceded by viral syndrome 1-3 weeks prior to onset (URI, diarrhea) or immunization (marginal increase from vaccinations, barely above baseline rate); viral exanthems in children, large viruses (CMV, EBV, HIV)
- **Campylobacter jejuni** most frequent antecedent infection
- Incident: 0.4-1.7 cases per 100,000 persons per year.
- History:
  - Earliest description by Wardrop and and Ollivier (1834).
  - Important landmarks: Landry's report (1859) acute, ascending, predominantly motor paralysis
  - **Guillain, Barré** and Strohl (1916) described benign polyneuritis with albuminocytologic dissociation.
  - Asbury and colleagues: perivascular, mononuclear, inflammatory infiltration of nerve roots and nerves.

AIDP Symptoms

- Earliest symptoms are paresthesias and numbness in fingers and toes.
- Symmetric weakness that evolves over days to weeks.
- Lower limbs usually first before spreading to the upper limbs, face, neck and trunk, thus the older term: *Landry Ascending Paralysis.*
- **5%** progress to total motor paralysis with respiratory failure w/in a few days.
- Severe cases can include ocular weakness and unreactive pupils.
- 50% with pain/aching in muscles (mainly hips, thighs, back) which can precede weakness.

AIDP Symptoms

- Sensory loss barely detectable early on. By end of the week, exam finds reduced touch-pressure-vibration.
- Reduced then absent DTRs consistent finding
- Facial weakness in 50% of patients, sometimes bilateral.
- Autonomic dysfunction-tachycardia (more common than bradycardia), facial flushing, fluctuating HTN or hypotension, loss of sweating or episodic profuse diaphoresis, urinary retention (15%).

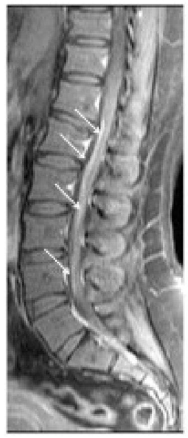
AIDP: CSF Findings

- CSF: Acellular or a few lymphocytes (10-50 cells/mm^3); cells decrease in number of days
- If there is a pleocytosis: think another cause for aseptic meningitis (HIV, sarcoid, Lyme, neoplastic infiltration)
- CSF: protein normal in first few days, then rises, and peaks at 4-6 weeks. (repeat CSF protein may be necessary if normal; rarely can be normal, especially in variants).
- **Albuminocytologic dissociation**
- **Glucose normal**

AIDP: Electrodiagnostic Findings

- Nerve Conduction abnormalities seen early and dependable indicator of GBS.
- Most seen:
  - Reduction in CMAP Amplitude
  - Slowed conduction velocities
  - Conduction block in motor nerves
  - Prolonged distal latencies, reduced distal amplitudes, prolonged or absent F-waves (indicating involvement or proximal nerve root involvement)
  - Delayed or absent H-reflex
  - Features of axonal damage correspond with poor recovery (positive EMG with denervation)

AIDP: Imaging Findings



- Can see gadolinium enhancement of the nerve roots, especially lumbosacral roots on MRI

AIDP: Variants

- Acute axonal form of AIDP
  - Abrupt, severe paralysis, with denervation due to axonal damage
  - 1/5th of cases associated with circulating antibody to G_{M1} ganglioside of the peripheral nerve.
- Complete ophthalmoplegia with ataxia (**Miller-Fisher Variant**) and isolated ophthalmoplegia
  - Associated with specific anti-neuronal antibody: **anti-GQ1b**

GBS: Treatment

- General Medical Care - 25% require mechanical ventilation
- Monitor respiratory, autonomic and motor function
- Hypotension - 10% of patients (IV saline and vasopressors)
- Hypertension-short-acting agents, example: IV clevidipine
- Electrolyte imbalance
- Precautions for Pulmonary embolus
- NG tube in some cases (paralytic ileus)
- SIADH
- Manage airway (bronchial secretions); possible tracheostomy for patients with rapid respiratory failure or rapidly quadriplegic

AIDP: Plasma Exchange and IVIg

- Consider Plasma exchange or IVIg if patient is unable to walk, reduced vital capacity or signs of oropharyngeal weakness.
- Plasma exchange: In patients treated within 2 weeks, there is halving of hospitalization, of the duration of mechanical ventilation, and time to reach independent ambulation. Less robust response if delayed > 2 weeks, but still better to treat than not.
- IVIg: equally effective to PE. Easier and safer to administer. Infrequent complications: renal failure, aseptic meningitis and proteinuria.
- 5-10% of patients with either PE or IVIg will relapse, and unable to predict who.
- Limited expectations for improvement in Axonal GBS.
- **No benefit from corticosteroids**

If the patient becomes unable to walk, shows a reduction in vital capacity, or signs of oropharyngeal weakness, plasma exchange or IVIg is instituted promptly. This typically occurs at the fifth to tenth day after the appearance of the first symptoms, but may be as early as 1 day or as late as 3 weeks.

GBS: Prognosis

- 3-5% do not survive
- In earliest stages, death due to acute cardiac arrest
- Majority recover with mild motor or sensory complaints in feet or legs
- 10% with pronounced residual disability
- Worse prognosis for axonal variants and patients with widespread weakness and prolonged mechanical ventilation.
- NCV/EMG evidence of axonal involvement: diffuse denervation with positive EMG and marked reduction in CMAP amplitudes have worse prognosis.
- Older adults recover more slowly
- 5-10% may have one or more recurrences